MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

A plane electromagnetic wave of frequency $50\mathrm{MHz}$ travels in free space. If the average energy densities in the electric field and magnetic field are $\mathrm{K_E}$ and $\mathrm{K_B}$ respectively, then the correct option in the following is

(A) $\mathrm{K_E} = \mathrm{K_B}$

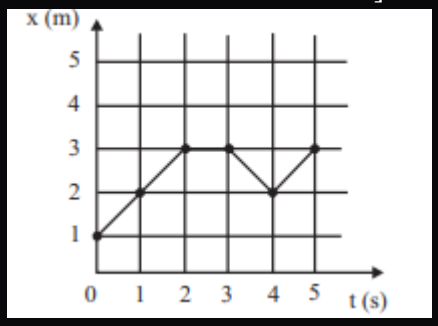
(B) $\mathrm{K_E} = \mathrm{K_B} = 0$

(C) $\mathrm{K_E} > \mathrm{K_B}$

(D) $\mathrm{K_E} < \mathrm{K_B}$

Q.2

A particle moves along a straight line along the x-axis. Its position (x) versus time (t) graph is shown in the figure [x in meters and t in seconds]. It's average speed during this motion is



(A) $0.4\ \mathrm{ms}^{-1}$

(B) $1.0\ \mathrm{ms}^{-1}$

(C) $0.8\ \mathrm{ms}^{-1}$

(D) $0.6\ \mathrm{ms}^{-1}$

Q.3

A open pipe of length l is vibrating in 3rd overtone with maximum amplitude A. The amplitude at a distance of $\frac{l}{16}$ from any open end is

(A) A

(B) 0

(C) $\frac{A}{\sqrt{2}}$

(D) $\frac{\sqrt{3}A}{2}$

Q.4

In an experiment of simple pendulum, the errors in the measurement of length of the pendulum (L) and time period (T) are 3% and 2%, respectively. The maximum percentage error in the value of $\frac{L}{T^2}$ is

(A) 5%

(B) 7%

(C) 8%

(D) 1%

Q.5

An object takes n times as much time as to slide down a 45° rough inclined plane as it takes to slide down a perfectly smooth inclined plane of the same inclination. The coefficient of kinetic friction between the object and the rough incline is given by

(A) $\left(1-\frac{1}{n^2}\right)$

(B) $\left(\frac{1}{1-n^2}\right)$

(C) $\sqrt{1-\frac{1}{n^2}}$

(D) $\sqrt{1+\frac{1}{n^2}}$

Q.6

Light of wavelength $488\ \text{nm}$ produced by an Argon laser is used in the photoelectric effect. When light from this spectral line is incident on the cathode, the stopping potential of the photoelectrons is $0.38\ \text{V}$. The work function of the cathode material is

(A) 2.16eV

(B) 216eV

(C) 21.6eV

(D) 0.216eV

Q.7

A body starts from rest and moves with uniform acceleration. If the distance travelled by it in the first 2 s is x_1 and in the next 2 s is x_2, then x_1 and x_2 are related as

(A) $x_1 = x_2$

(B) $x_1 = 2x_2$

(C) $2x_1 = x_2$

(D) $3x_1 = x_2$

Q.8

In a RC circuit, where R is resistance and C is capacitance which of the following has the dimension of time.

(A) $\frac{R}{C}$

(B) $\frac{C}{R}$

(C) $\sqrt{RC}$

(D) RC

Q.9

All gases deviate from gas laws at

(A) low pressure and high temperature

(B) high pressure and low temperature

(C) low pressure and low temperature

(D) high pressure and high temperature

Q.10

A bar magnet having magnetic moment M is bent as an arc. Its magnetic moment

(A) decreases

(B) increases

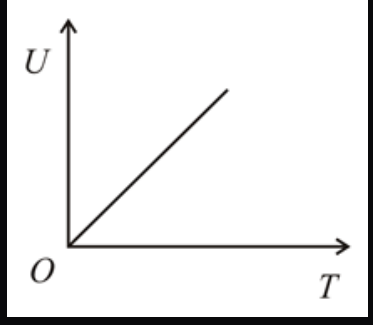
(C) does not change

(D) may increase or decrease

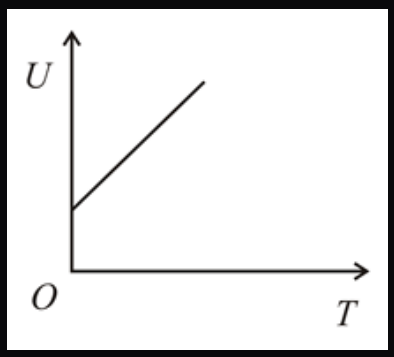
Q.11

Which one of the graphs below best illustrates the relationship between internal energy U of an ideal gas and temperature T of the gas in K?

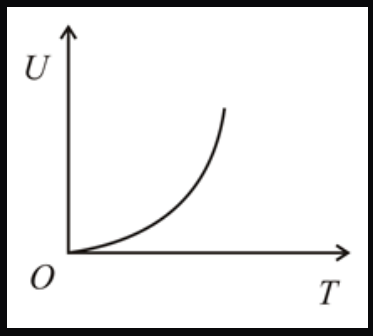
(A)



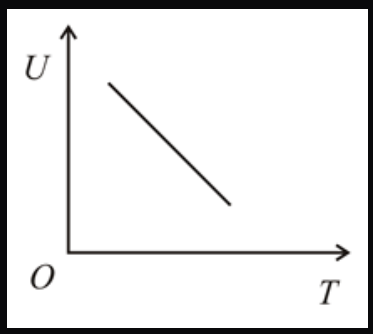
(B)



(C)



(D)



Q.12

The energy of a photon is equal to the kinetic energy of a proton. If λ_1 is the de-Broglie wavelength of a proton, λ_2 the wavelength associated with the photon and if the energy of the photon is E, then (λ_1/λ_2) is proportional to

(A) E^4

(B) $E^{1/2}$

(C) E^2

(D) E

Q.13

The truth table given below corresponds to logic gate.

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

(A) NAND

(B) OR

(C) AND

(D) XOR

Q.14

Two slabs A and B of different materials but of the same thickness are joined end to end to form a composite slab. The thermal conductivities of A and B are k_1 and k_2 respectively. A steady temperature difference of $12^\circ\mathrm{C}$ is maintained across the composite slab. If $k_1 = \frac{k_2}{2}$, the temperature difference across slab A is

(A) $4^\circ\mathrm{C}$

(B) $6^\circ\mathrm{C}$

(C) $8^\circ\mathrm{C}$

(D) $10^\circ\mathrm{C}$

Q.15

A wire of length l is bent into a circular loop of radius R and carries a current I. The magnetic field at the centre of the loop is B. The same wire is now bent into a double loop of equal radii. If both loops carry the same current I and it is in the same direction, the magnetic field at the centre of the double loop will be

(A) Zero

(B) 2 B

(C) 4 B

(D) 8 B

Q.16

A circular freeway entrance and exit are commonly banked to control a moving car at $14\ \mathrm{m/s}$. To design similar ramp for $28\ \mathrm{m/s}$ one should

(A) increase the radius by factor 2

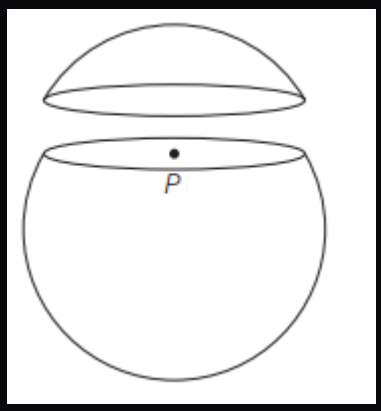
(B) increase the radius by factor 4

(C) decrease the radius by factor 4

(D) decrease the radius by factor 2

Q.17

A spherical shell is cut into two pieces along a chord as shown in the figure. P is a point on the plane of the chord. The gravitational field at P due to the upper part is I_1 and that due to the lower part is I_2. What is the relation between them?



(A) $I_1 > I_2$

(B) $I_2 > I_1$

(C) $I_1 = I_2$

(D) No definite relation

Q.18

A machine which is 70% efficient raises a $10\ \text{kg}$ body through a certain distance and spends $100\ \text{J}$ energy. The body is then released. On reaching the ground, the kinetic energy of the body will be

(A) 0

(B) $70\ \text{J}$

(C) $50\ \text{J}$

(D) $35\ \text{J}$

Q.19

An infinite number of electric charges each equal to 5 nano-coulomb (magnitude) are placed along X-axis at $x = 1\ \text{cm}, x = 2\ \text{cm}, x = 4\ \text{cm}, x = 8\ \text{cm}....$ and so on. In this set up if the consecutive charges have opposite sign, then the electric field in newton/coulomb at $x = 0$ is :
$\left(\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ \text{N} - \text{m}^2/\text{C}^2\right)$

(A) 12×10^4

(B) 24×10^4

(C) 36×10^4

(D) 48×10^4

Q.20

A $2.0\ \text{cm}$ object is placed $15\ \text{cm}$ in front of a concave mirror of focal length $10\ \text{cm}$. What is the size and nature of the image?

(A) $4\ \text{cm}$. real

(B) $4\ \text{cm}$, virtual

(C) $1.0\ \text{cm}$, real

(D) None

Q.21

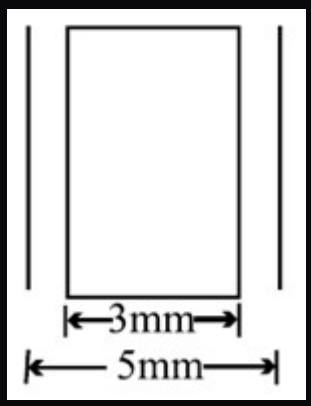
In Young's double slit experiment, an interference pattern is obtained on a screen by a light of wavelength $6000\mathring{A}$ coming from the coherent sources S_1 and S_2. At certain point P on the screen third dark fringe is formed. Then, the path difference $S_1P - S_2P$ in microns is K, find 4K.

Q.22

An electron revolves in a circle of radius $0.4\mathring{A}$ with a speed of $10^6\ \text{m/s}$ in a hydrogen atom. The magnetic field produced at the centre of the orbit due to the motion of the electron (in Tesla) is : $\left[\mu_0 = 4\pi \times 10^{-7}\text{H/m}\right.$, charge on the electron $\left. = 1.6 \times 10^{-19}\text{C}\right]$

Q.23

Separation between the plates of a parallel plate capacitor is $5\ \text{mm}$. This capacitor, having air as the dielectric medium between the plates, is charged to a potential difference $25\ \text{V}$ using a battery. The battery is then disconnected and a dielectric slab of thickness $3\ \text{mm}$ and dielectric constant $K = 10$ is placed between the plates as shown. Potential difference between the plates after the dielectric slab has been introduced is Y, find 2Y



Q.24

Three identical spheres each of diameter $2\sqrt{3}\ \text{m}$ are kept on a horizontal surface such that each sphere touches the other two spheres. If one of the sphere is removed, then the shift in the position of the centre of mass of the system is (in m)

Q.25

A solenoid of length 60 cm with 15 turns per cm and area of cross section 4×10^{-3} m^2 completely surrounds another co-axial solenoid of same length and area of cross-section 2×10^{-3} m^2 with 40 turns per cm. Mutual inductance of the system is _____ mH.

Q.26

In which one of the following pairs the two species have identical shape, but differ in hybridisation?

(A) $I_3^-, BeCl_2$

(B) NH_3, BF_3

(C) XeF_2, I_3^-

(D) NH_4^-, SF_4

Q.27

A primary alcohol was reacted with pyridinium chlorochromate (PCC), which resulted in a product P. The product P on treatment with ammonical silver nitrate solution produces.

(A) Anhydride of carboxylic acid

(B) Aldehyde

(C) Amide

(D) Carboxylate anion

Q.28

Two compounds form an ideal solution at room temperature. Which of the following are correct for this ideal solution?

(A) $\Delta G = (+ve)$

(B) $\Delta S = (+$ ve) surrounding

(C) $\Delta S = (+ve)$ system

(D) $\Delta_{max} H = 0$

(A) C, D

(B) B, C, D

(C) B, C

(D) A, D

Q.29

Which statement is incorrect

(A) Phenol is a weak acid

(B) Phenol is an aromatic compound

(C) Phenol liberates CO_2 from Na_2CO_3 soln

(D) Phenol is soluble in $NaOH$

Q.30

Which of the following statements are correct?
i. $[Mn(CN)_6]^{3-}$, $[MnCl_6]^{3-}$ both are paramagnetic
ii. Both $[NiCl_4]^{2-}$, $[Ni(CO)_4]$ are tetrahedral and diamagnetic
iii. $[Co\ (NH_3)_6]^{3+}$ is inner orbital complex, but $[Ni\ (NH_3)_6]^{2+}$ is outer orbital complex

(A) i, ii only

(B) i, iii only

(C) ii, iii only

(D) i, ii, iii

Q.31

Which one of the following gives Prussian blue colour?

(A) $Fe_2\ [Fe(CN)_6]$

(B) $Na_4\ [Fe(CN)_6]$

(C) $Fe_3\ [Fe(CN)_6]_3$

(D) $Fe_4\ [Fe(CN)_6]_3$

Q.32

In thin layer chromatography, a substance moved by a distance of $4\ cm$, where as the solvent moved by a distance of $5\ cm$. Then the value of retardation factor (R_f) is

(A) 0.8

(B) 1.0

(C) 9.0

(D) 1.25

Q.33

Given below are two statements
Assertion (A): Hydrolysis of DNA does not form equal number of A and T; G and C
Reason (R): In DNA adenine forms hydrogen bonds with thymine and cytosine forms hydrogen bonds with guanine
The correct answer is

(A) Both A and R are correct and R is the correct explanation of A

(B) Both A and R are correct but R is not the correct explanation of $\dot{A}$

(C) A is correct but R is incorrect

(D) A is incorrect but R is correct

Q.34

The species Ar, K^+ and Ca^{2+} contain the same number of electrons. In which order do their radii increase?

(A) $\mathrm{Ar} < \mathrm{K}^+ < \mathrm{Ca}^{2+}$

(B) $\mathrm{Ca}^{2+} < \mathrm{K}^+ < \mathrm{Ar}$

(C) $\mathrm{K}^+ < \mathrm{Ar} < \mathrm{Ca}^{2+}$

(D) $\mathrm{Ca}^{2+} < \mathrm{Ar} < \mathrm{K}^+$

Q.35

Which of the following is the most basic oxide ?

(A) SeO_2

(B) Al_2O_3

(C) Sb_2O_3

(D) Bi_2O_3

Q.36

Steam distillation process cannot be used for purifying which of the following?

(A) Aniline

(B) p-nitrophenol

(C) Toluene

(D) Nitrobenzene

Q.37

The atomic numbers of elements A, B, C and D are $Z-1, Z, Z+1$ and $Z+2$, respectively. If ' B ' is a noble gas, choose the correct answers from the following statements

(1) ' A ' has higher electron affinity

(2) ' C ' exists in +2 oxidation state

(3) ' D ' is an alkaline earth metal

(A) (1) and (2)

(B) (2) and (3)

(C) (1) and (3)

(D) $(1), (2)$ and (3)

Q.38

When compound X is oxidised by acidified potassium dichromate, compound Y is formed. Compound Y on reduction with $LiAlH_4$ gives $X.X$ and Y respectively are :

(A) C_2H_5OH, CH_3COOH

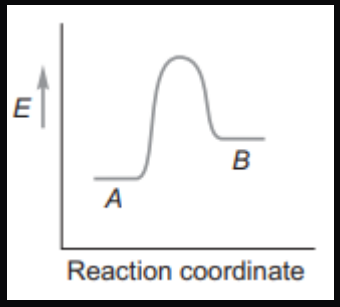
(B) CH_3COCH_3, CH_3COOH

(C) C_2H_5OH, CH_3COCH_3

(D) CH_3CHO, CH_3COCH_3

Q.39

For a reversible reaction $A \rightleftharpoons B$, which one of the following statements is wrong from the given energy profile diagram?



(A) Activation energy of forward reaction is greater than backward reaction

(B) The forward reaction is endothermic

(C) The threshold energy is less than that of activation energy

(D) The energy of activation of forward reaction is equal to the sum of heat of reaction and the energy of activation of backward reaction

Q.40

If the resistance of $0.1M$ KCl solution in a conductance cell is 300Ω and conductivity is $0.013 Scm^{-1}$, then the value of cell constant is

(A) $3.9\ cm^{-1}$

(B) $39\ m^{-1}$

(C) $3.9\ m^{-1}$

(D) $0.39\ cm^{-1}$

Q.41

The equilibrium constant for the reaction
$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$ is 5×10^{-2} atm.
The equilibrium constant of the reaction
$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$ would be

(A) $100\ atm$

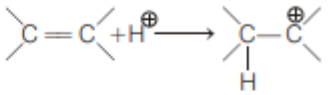
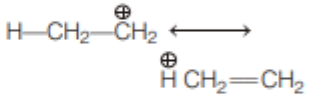
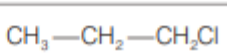
(B) $200\ atm$

(C) 4×10^2 atm

(D) 6.25×10^4 atm

Q.42

Match the following.

List-I		List-II
(A) Resonance	(I)	
(B) Inductive effect	(II)	
(C) Electromeric effect	(III)	
(D) Hyperconjugation	(IV)	
	(V)	

The correct answer is

A B C D

(A) II I IV III

(B) III V I II

(C) I III II V

(D) III II I IV

Q.43

An organic compound X gives a red precipitate on heating with Fehling's solution. Which one of the following reactions yields X as a major product?

(A) $\mathrm{HCHO} \xrightarrow[\text{(ii) } \mathrm{H_2O}]{\text{(i) } \mathrm{CH_3MgI}}$

(B) $\mathrm{C_2H_5Br + AgOH} \xrightarrow{\Delta}$

(C) $\mathrm{2C_2H_5Br + Ag_2O} \xrightarrow{\Delta}$

(D) $\mathrm{C_2H_2 + H_2O} \xrightarrow[1\%\mathrm{HgSO_4}\,60^\circ\mathrm{C}]{40\%\mathrm{H_2SO_4}}$

Q.44

Using the standard electrode potentials given below identify the correct statements from the following.

$$\mathrm{Fe^{2+} + 2e^- \longrightarrow Fe;\ } E^\circ = -0.44\ \mathrm{V}$$

$$\mathrm{Cu^{2+} + 2e^- \longrightarrow Cu;\ } E^\circ = +0.34\ \mathrm{V}$$

$$\mathrm{Ag^{+} + e^- \longrightarrow Ag;\ } E^\circ = +0.80\ \mathrm{V}$$

(i) Copper can displace iron from $FeSO_4$ solution.
(ii) Iron can displace copper from $CuSO_4$ solution.
(iii) Silver can displace copper from $CuSO_4$ solution.
(iv) Iron can displace silver from $AgNO_3$ solution.

(A) (i), (ii)

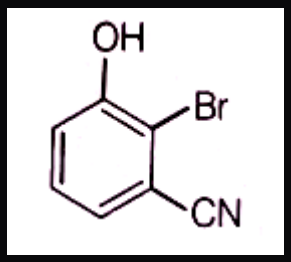
(B) (ii), (iii)

(C) (ii), (iv)

(D) (i), (iv)

Q.45

The IUPAC name of the following compound is



(A) 1-bromo-2-hydroxybenzonitrile

(B) 2-bromo-3-cyanophenol

(C) 2-bromo-3-hydroxy benzonitrile

(D) 3-hydroxy-2-brómobenzonitrile

Q.46

How many among the given species have a bond order of 0.5 ?
$H_2^+, He_2^+, He_2^-, B_2^+, F_2^-, Be_2^{2-}$

Q.47

From the list given below, the number of lanthanides which exhibit +4 state in their oxides is
Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy

Q.48

A metal plate of dimension of $(1 \times 2\ cm^2)$ has to be coated on both the sides by Cu metal. How long does it take to deposit Cu of $0.01\ cm$ thickness, if $1.5\ A$ current is used? [Electrochemical equivalence of Cu is $0.0003\ g/C$ and the density of Cu is $9\ g/cm^3$] Mark answer in seconds

Q.49

Nitrobenzene on reduction using zinc in alkaline medium results in X. The number of $sigma(\sigma)$ and pi (π) bonds in X is A and B respectively. Find A+2B.

Q.50

The number of electron that can be present in sub-shells having m_s value of $\frac{-1}{2}$, for n up to 3.

Q.51

Consider all rectangles lying in the region
$\left\{(x, y) \in \mathbb{R} \times \mathbb{R}: 0 \leq x \leq \frac{\pi}{2} \text{ and } 0 \leq y \leq 2 \sin (2 x)\right\}$
and having one side on the x-axis. The area of the rectangle which has the maximum perimeter among all such rectangles, is

(A) $\frac{3 \pi}{2}$

(B) π

(C) $\frac{\pi}{2 \sqrt{3}}$

(D) $\frac{\pi \sqrt{3}}{2}$

Q.52

If the function $f(\mathrm{x})=2+\mathrm{x}^2-\mathrm{e}^{-\mathrm{x}}$ and $\mathrm{g}(\mathrm{x})=f^{-1}(\mathrm{x})$, then the value of $\frac{\left|\mathrm{g}^{\prime}(1)\right|}{4}$ equals

(A) 0.5

(B) 1

(C) 1.25

(D) 0.25

Q.53

The sum of two natural numbers n_1 and n_2 is known to be equal to 100. The probability that their product being greater than 1600 , is equal to

(A) $\frac{20}{33}$

(B) $\frac{58}{99}$

(C) $\frac{13}{33}$

(D) $\frac{59}{99}$

Q.54

Let $T_n=3 \mathrm{n}-2$ and $T_m^{\prime}=7 \mathrm{~m}+2$ be two sequences each having 2004 terms. The number of common terms to the two sequences is-

(A) 283

(B) 284

(C) 285

(D) 286

Q.55

If $\left(x^2-5x+4\right)\left(y^2+y+1\right)<2y$ for all real y, then x belongs to the interval $(2, b)$, then b can be

(A) 3

(B) 4

(C) 5

(D) 6

Q.56

The value of the limit $\lim_{x \to \frac{\pi}{2}} \frac{4\sqrt{2}(\sin 3x+\sin x)}{\left(2 \sin 2x \sin \frac{3x}{2}+\cos \frac{5x}{2}\right)-\left(\sqrt{2}+\sqrt{2} \cos 2x+\cos \frac{3x}{2}\right)}$ is _____

(A) 6

(B) 4

(C) 2

(D) 8

Q.57

The number of matrices of A for which the system of linear equations $A\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has a unique solution, is

(A) less than 4

(B) atleast 4 but less than 7

(C) atleast 7 but less than 10

(D) atleast 10

Q.58

A, B, C are vertices of a triangle with right angle at A and $P(-4,0); Q(0,6)$ are two given points. If the ratio of distances from each vertex to P, to that of Q is $2:3$, then the centroid of $\triangle ABC$ lies on a circle with radius equal to

(A) $\frac{4\sqrt{13}}{5}$ units

(B) 4 units

(C) $\frac{8\sqrt{13}}{5}$ units

(D) 8 units

Q.59

If $\left(1+ax+bx^2\right)^4=a_0+a_1x+a_2x^2+\ldots+a_8x^8$ where $a,b,a_0,a_1,\ldots.a_8\in R$ and $a_0+a_1+a_2\neq0$ and $\begin{vmatrix}a_0&a_1&a_2\\a_1&a_2&a_0\\a_2&a_0&a_1\end{vmatrix}=0$, then value of $5\left(\frac{a}{b}\right)$.

(A) 8

(B) 9

(C) 10

(D) 6

Q.60

Let $\mathrm{y}=f(\mathrm{x})$ satisfies the differential equation, $\mathrm{xy}(1+\mathrm{y})\mathrm{dx}=\mathrm{dy}$. If $f(0)=1$ and $\mathrm{k}f(2)=(1+f(2))\mathrm{e}^2$, where $\mathrm{k}\in\mathrm{N}$, then k is equal to
[Note : e denotes Napier's constant]

(A) 1

(B) 3

(C) 2

(D) 4

Q.61

If f and g are two real valued differentiable functions such that $f(x)=x^2+g'(0)x+g''(1)$ and $g(y)=y^2+f''(y)+f'(y)-2$ then $\mathrm{f}(3)+\frac{1}{\mathrm{g}(2)}$ is equal to

(A) 21.1

(B) 17.1

(C) 15.1

(D) 6.1

Q.62

Let A be any point on the ellipse $x^2+16y^2=4$ with foci S_1 and S_2 then the maximum value of $\mathrm{AS}_1\cdot\mathrm{AS}_2$ is

(A) 8

(B) 4

(C) 12

(D) 6

Q.63

If the equation $\left(x^2+2\right)^2-(a+4)\left(x^4+3x^2+2\right)+(a+3)\left(x^2+1\right)^2=0$ has atleast one real solution, then integral value of ' a ' can be-

(A) -2

(B) -1

(C) 0

(D) 1

Q.64

If $\mathrm{f(x)} = \lim_{a \to x} \left(\frac{e^{a^2+|x|-x^2} - e^{|x|}}{(a+x)\sin(a-x)} \right); \forall x \in R$ then number of roots of the equation $f(x)\left(\left|x^2-1\right|\right) = 1$ is

(A) 2

(B) 3

(C) 4

(D) 5

Q.65

The reciprocal of the distance between two points, one on each of the lines $\frac{x-2}{3} = \frac{y-4}{2} = \frac{z-5}{5}$ and $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$,
Then which of the following is FALSE.

(A) can be 8

(B) can be less than $\sqrt{75}$

(C) cannot be greater than 78

(D) can be $\sqrt{80}$

Q.66

If $\mathrm{x} = \alpha$ satisfies the equation $\frac{\sin^{-1}\mathrm{x}^2 + \cos^{-1}\mathrm{x}}{\cos^{-1}\mathrm{x}^2 + \sin^{-1}\mathrm{x}} = -3$, then the value of $\frac{\sec^{-1}\alpha - \tan^{-1}\alpha}{\cot^{-1}\alpha - \mathrm{cosec}^{-1}\alpha}$ is

(A) 1

(B) 1.5

(C) 2

(D) 2.5

Q.67

A line L cuts the lines AB, AC and AD of a parallelogram $ABCD$ at points B_1, C_1 and D_1 respectively. If $\overrightarrow{\mathrm{AB_1}} = \lambda_1\overrightarrow{\mathrm{AB}}, \overrightarrow{\mathrm{AD_1}} = \lambda_2\overrightarrow{\mathrm{AD}}$ and $\overrightarrow{\mathrm{AC_1}} = \lambda_3\overrightarrow{\mathrm{AC}}$ then $\frac{1}{\lambda_3} =$

(A) $\frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

(B) $\frac{1}{\lambda_1} - \frac{1}{\lambda_2}$

(C) $-\lambda_1 + \lambda_2$

(D) $\lambda_1 + \lambda_2$

Q.68

If the chord of the circle $x^2 + y^2 - 4x - 2y - c = 0$ is trisected at the points $\left(\frac{1}{3}, \frac{1}{3}\right)$ and $\left(\frac{8}{3}, \frac{8}{3}\right)$ then :

(A) Length of the chord $= 7$

(B) $\mathrm{c} = 20$

(C) Radius of circle $= 6$

(D) $\mathrm{c} = 25$

Q.69

I alternatively toss a fair coin and a fair die until I, either toss a head or throw a '2'. If I toss the coin first, the probability that I throw a '2' before I toss a head, is

(A) $1/7$

(B) $7/12$

(C) $5/12$

(D) $5/7$

Q.70

Let $\mathrm{y} = f(\mathrm{x})$ is a curve such that slope of tangent at any point is equal to sum of ratio of ordinate and abscissa of that point and square of its abscissa. If the curve passes through $(1, -2)$, then $2\int_0^1 f(\mathrm{x})\mathrm{dx}$ is equal to

(A) 2.5

(B) -2.5

(C) 2.25

(D) -2.25

Q.71

The value of $\lim_{t\to 0} \frac{\frac{1}{t^3}\int_0^t \left(\frac{1-\cos(3x)}{x^2+3}\right)dx}{\int_{-\frac{1}{2}}^{\frac{1}{2}} \sin\left(\tan^{-1}\left(\frac{x^2}{\sqrt{1-x^4}}\right)\right)dx}$ is equal to

Q.72

Let $|\vec{a}| = \frac{1}{2}; |\vec{b}| = \frac{1}{3}; |\vec{c}| = \frac{1}{6}; |\vec{d}| = 1$ such that $\vec{a} + \vec{b} + \vec{c} + \vec{d} = 0$ as well as $\vec{a} \wedge \vec{b} = \theta_1$ and $\vec{c} \wedge \vec{d} = \theta_2$, then $(\cos\theta_1 - \cos\theta_2)$ is

Q.73

The direction cosines (l, m, n) of two lines satisfy $3l + m + 5n = 0, 6mn - 2nl + 5lm = 0$. If ' θ ' is the angle between those two lines then the value of $\left[\frac{6\tan^2\theta - 5\sec\theta}{10\pi}\right]$ where [.] denotes greatest integer function is

Q.74

Let $F(z) = \frac{z+i}{z-i}$ for all complex numbers $z \neq i$, and let $z_n = F(z_{n-1})$ for all positive integers n. Given that $z_0 = \frac{1}{137} + i$ and $z_{2002} = a + bi$, where a and b are real numbers, find unit digit of $a + b$.

Q.75

There is a group of 265 persons who like either singing or dancing or painting. In this group 200 like singing, 110 like dancing and 55 like painting. If 60 persons like both singing and dancing, 30 like both singing and painting and 10 hike all three activities, then the number of persons who like only dancing and painting is

Answers & Solutions

Q.1 Answer:

—

Solution:

$$\mathrm{K_E} = \tfrac{1}{2}\varepsilon_0 \mathrm{E_0^2} = \tfrac{1}{2}\varepsilon_0\, \mathrm{B_0^2 C^2}$$
$$= \tfrac{1}{2}\varepsilon_0 B_0^2 \times \tfrac{1}{\mu_0\varepsilon_0} = \tfrac{B_0^2}{2\mu_0} = K_B$$

Q.2 Answer:

$0.8\ \mathrm{ms^{-1}}$

Solution:

$$\text{Average speed } = \frac{\text{Distance travelled}}{\text{time}}$$
$$= \frac{4\ \mathrm{m}}{5\ \mathrm{s}} = 0.8\ \mathrm{m/s}$$

Q.3 Answer:

$\frac{A}{\sqrt{2}}$

Solution:

For open organ pipe, wavelength of nth overtone

$$\lambda_n = \frac{2l}{n+1} \quad (n = 3)$$
$$\lambda = \frac{2l}{4} = \frac{l}{2}$$

As pipe is open, so antinode (maximum amplitude) will form at open ends.

At $l = 0$, amplitude $= A$

At $\frac{l}{16}$ or $\frac{\lambda}{8}$ distance, amplitude, $R = A\cos\phi$

where, $\phi =$ phase angle.

$$\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{8} = \frac{\pi}{4} = 45^\circ$$

$$R = A\cos 45^\circ$$

$$R = \frac{A}{\sqrt{2}}$$

Q.4 Answer:

7%

Solution:

Maximum percentage in the value of $\frac{L}{T^3}$

$$= \frac{\Delta L}{L} \times 100\% + 2\frac{\Delta T}{T} \times 100\%$$

$$= 3 + 2 \times 2 = 7\%$$

Q.5 Answer:

$\left(1 - \frac{1}{n^2}\right)$

Solution:

The coefficient of kinetic friction between the object and the rough incline

$$\mu = \tan\theta\left(1 - \tfrac{1}{n^2}\right)$$

$$\mu = \tan 45^\circ\left(1 - \tfrac{1}{n^2}\right)$$

$$= 1 - \tfrac{1}{n^2}$$

Q.6 Answer:

2.16eV

Solution:

Given, light of wavelength, $\lambda = 488\ \text{nm}$ and stopping potential, of photoelectrons, $V = 0.38\ \text{V}$ As, energy of photon

$$E = \frac{hc}{\lambda} = \frac{1250}{488}\text{eV}$$

So, $E = 2.56\text{eV}$

$\because$ Maximum kinetic energy of a photo-electron

$$\text{KE}_{\max} = eV_0$$

Where, $V_0 =$ stopping potential

$$\therefore \quad \text{KE}_{\max} = eV_0 = 0.38\text{eV}$$

From Einstein's photoelectric equation,

$$E = \text{KE}_{\max} + W_0$$

$$\Rightarrow \quad W_0 = E - \text{KE}_{\max}$$

$$= 2.56 - 0.38 = 2.18 \approx 2.16\text{eV}$$

Hence, the work function of cathode material is 2.16eV.

So, the correct option is (a).

Q.7 Answer:

$3x_1 = x_2$

Solution:

If a be the acceleration of the body, then distance travelled by the body in 2 s is given as,

$$x_1 = ut + \frac{1}{2}at^2 = 0 \times 2 + \frac{1}{2}a \times (2)^2$$

$$x_1 = 2a$$

Velocity of body at the end of 2 s is given as,

$$v = u + at = 0 + a \times 2$$

$$v = 2a$$

Distance travelled by the body in next 2 s is given as,

$$x_2 = vt + \frac{1}{2}at^2$$

$$= 2a \times 2 + \frac{1}{2}a \times (2)^2 \text{ [from Eq. (ii)]}$$

$$= 4a + 2a$$

$$x_2 = 6a = 3 \times 2a$$

$$\Rightarrow \quad x_2 = 3 \times x_1 \quad \text{[from Eq. (i)]}$$

$$\Rightarrow \quad x_2 = 3x_1$$

Q.8 Answer:

RC

Solution:

The current in DC ciruit containing resistance and capacitance is given by,

$i = i_0\left(1 - e^{-\frac{t}{RC}}\right)$

As the power of e should be dimensionless, the dimensions of RC should be equal to that of time.

Q.9 Answer:

high pressure and low temperature

Solution:

At high pressure, the volume occupied by gas molecules does not approach to zero and at low temperature, the contribution of intermolecular forces becomes significant. Hence, at high pressure and low temperature, all gases deviate from gas laws.

Q.10 Answer:

decreases

Solution:

Magnetic moment of a bar magnet,

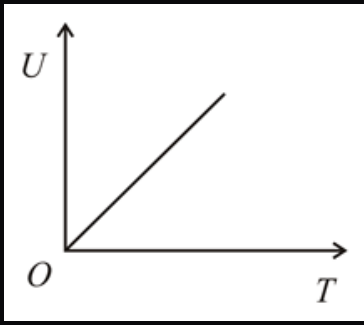
$\mathrm{M} = m(2l)$

$=$ pole strength $\times$ distance between poles

When bar magnet is bent as an arc distance between poles decreases whereas pole strength remains same.

So, its magnetic moment also, decreases.

Q.11 Answer:



Solution:

$$U = nC_V \;\; T = \frac{f}{2}nRT \propto T$$

Q.12 Answer:

$E^{1/2}$

Solution:

We know that, de-Broglie wavelength of photon

$E = h\mathrm{v}$

But we know that also

$$E = \frac{1}{2}mv^2$$

$$\therefore \quad 2E = \frac{P^2}{m}$$

$$\because \quad P = mv$$

$$\frac{P^2}{m} = mv^2$$

$$P = \sqrt{2mE}$$

(for proton)

According to the question

$\lambda_1 = \frac{h}{P}$

(de-Broglie of matter waves)

and $\lambda_2 = \frac{hc}{E}$

From the Eqs. (ii) and (iii), we get

$$\frac{\lambda_1}{\lambda_2} = \frac{h/P}{hc/E} = \frac{E}{P_C}$$

$$= \frac{E}{\sqrt{2mE} \cdot C} = \frac{\sqrt{E}}{\sqrt{2m} \cdot C}$$

So, $\frac{\lambda_1}{\lambda_2} = \frac{\sqrt{E}}{\sqrt{2m} \cdot c}$

$\frac{\lambda_1}{\lambda_2} \propto \sqrt{E}$

$\Rightarrow \quad \lambda_1 : \lambda_2 = E^{1/2}$

Q.13 Answer:

OR

Solution:

The truth table given below corresponds to logic gate.

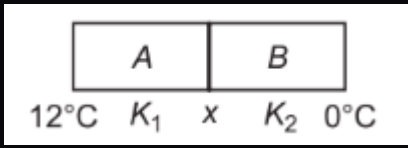
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Q.14 Answer:

$8^\circ C$

Solution:

The given situation can be shown as



Rate of flow of heat will be equal in both the slabs

$\therefore (12 - x)K_1 = K_2(x - 0)$

$12 - x = 2x \quad (\because K_1 = \frac{K_2}{2})$

$x = 4^\circ C$

The temperature difference across slab

$A = (12 - x) = (12 - 4)$

$= 8^\circ C$

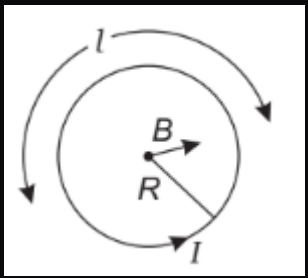
Q.15 Answer:

4 B

Solution:

Magnetic field at the centre of the loop

$$B = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot 2\pi R}{R^2} \qquad \ldots(i)$$



For the wire which is looped double let radius becomes r Then,

$$\frac{l}{2} = 2\pi r$$

$$\text{or} \quad \frac{l}{4\pi} = (r)$$

$$\therefore \quad B' = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot 2\pi r \times 2}{r^2}$$

$$\text{or} \quad B' = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot \frac{l}{2} \cdot 2}{\left(\frac{l}{4\pi}\right)^2}$$

$$\text{or } B' = \frac{\mu_0}{4\pi} \cdot \frac{Il \times 16\pi^2}{l^2} \quad \ldots(ii)$$

$$\text{Now, } B = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot l}{\left(\frac{l}{2\pi}\right)^2} \left[R = \frac{l}{2\pi}\right] \quad \ldots(iii)$$

Dividing Eq. (ii) by Eq. (iii), we get

$$\frac{B'}{B} = \frac{\frac{\mu_0}{4\pi} \cdot \frac{I \cdot l \cdot 16\pi^2}{l^2}}{\frac{\mu_0}{4\pi} \cdot \frac{Il \cdot 4\pi^2}{l^2}}$$

$$\text{or} \quad \frac{B'}{B} = 4$$

$$\text{or} \quad B' = 4B$$

Q.16 Answer:

increase the radius by factor 4

Solution:

Given, $v_1 = 14 \text{ m/s}$ **and** $v_2 = 28 \text{ m/s}$

As we know that, $\tan\theta = \frac{v^2}{rg} \Rightarrow r = \frac{v^2}{g\tan\theta}$

In the first case, $r_1 = \frac{v_1^2}{g\tan\theta} = \frac{(14)^2}{g\tan\theta} \ldots$ **(i)**

In the second case,

$r_2 = \frac{(28)^2}{g\tan\theta}$ **[for similar ramp remains same]**

$r_2 = \frac{(14\times 2)^2}{g\tan\theta} = 4\frac{(14)^2}{g\tan\theta} \Rightarrow r_2 = 4r_1 \quad [\text{using Eq. (i)}]$

Hence, radius should be increased by factor 4 .

Q.17 Answer:

$I_1 = I_2$

Solution:

At point P**, we have** $I_1 - I_2 = 0$

Because gravitational field inside a shell is zero.

Hence, $I_1 = I_2$

Q.18 Answer:

70 J

Solution:

Energy spent by the machine to raise the body of mass $10\ \text{kg}$ at certain height, is $E = 100\ \text{J}$

Since, machine is only 70% efficient.

Hence, energy used by machine,

$$E' = 70\% \text{ of } E$$

$$= \frac{70}{100} \times 100 = 70\ \text{J}$$

When body is released, then whole energy spent by machine is converted into kinetic energy on reaching the ground.

$\therefore$ Kinetic energy, $K = E' = 70\ \text{J}$

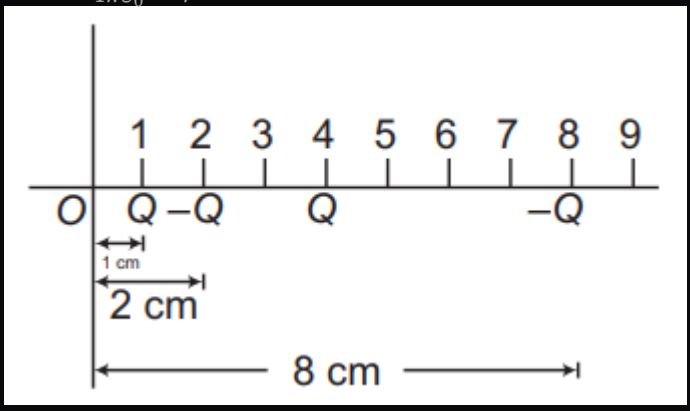
Q.19 Answer:

36×10^4

Solution:

Electric field intensity due to a point charge.

$$E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{Q}{r^2}$$



The consecutive charges are of opposite signs.

$\therefore$ Net electric field at $x = 0$, is

$$E = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q}{r_1^2} - \frac{Q}{r_2^2} + \frac{Q}{r_3^2} - \frac{Q}{r_4^2} \ldots..\infty\right]$$

$$= \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_1^2} - \frac{1}{r_2^2} + \frac{1}{r_3^2} - \frac{1}{r_4^2} + \ldots.\infty\right]$$

$$= 9 \times 10^9 \times 5 \times 10^{-9}$$

$$\left[\frac{1}{(1\times10^{-2})^2} - \frac{1}{(2\times10^{-2})^2} + \frac{1}{(4\times10^{-2})^2} + \ldots\infty\right]$$

$$= \frac{45}{10^{-4}}\left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{4^2} - \frac{1}{8^2} + \ldots.\infty\right]$$

$$= 45 \times 10^4\left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{4^2} - \frac{1}{8^2} \ldots\infty\right]$$

$$\left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{4^2} - \frac{1}{8^2} \ldots\infty\right] = \frac{\frac{1}{1^2}}{1-\left(\frac{-1}{2}\right)^2}$$

[Sum of infinite GP, $S_\infty = \frac{a}{1-r}$]

$= \frac{1}{1+\frac{1}{4}} = \frac{4}{5}$

$\therefore \quad E = 45 \times 10^4 \times \frac{4}{5} = 36 \times 10^4 \text{ N/C}$

Q.20 Answer:

4 cm. real

Solution:

$\frac{1}{\text{v}} - \frac{1}{15} = \frac{1}{-10} \Rightarrow \text{v} = -30 \text{ cm}$

$m = \frac{-30}{-15} = 2$, **image size** $= 4$ cm

Q.21 Answer:

—

Solution:

$\lambda = 6000\mathring{A} = 6 \times 10^{-7} \text{ m}$

Path difference for dark fringe;

$\Delta x = (2n+1)\lambda/2$

For third dark fringe, $n = 2$

$\therefore \quad \Delta x = (2 \times 2 + 1) \times \frac{6 \times 10^{-7}}{2}$

$= \frac{5 \times 6 \times 10^{-7}}{2}$

$= 15 \times 10^{-7}$

$= 1.5 \times 10^{-6} \text{ m}$

$= 1.5 \mu\text{m}$

Q.22 Answer:

—

Solution:

$r = 0.4\tilde{A} \ldots, = 0.4 \times 10^{-10} \text{ m}, v = 10^6 \text{ m/s}$

The moving electron will constitute an electric current

Electric current $i = \frac{q}{t}$

$$i = \frac{q}{\frac{2\pi r}{v}} = \frac{qv}{2\pi r}$$

$$i = \frac{1.6 \times 10^{-19} \times 10^6}{2\pi \times 0.4 \times 10^{-10}}$$

$$= \frac{1.6 \times 10^{-3}}{0.8\pi}$$

Magnetic produced at the centre

$$B = \frac{\mu_0 i}{2r}$$

$$B = \frac{4\pi \times 10^{-7}}{2 \times 0.4 \times 10^{-10}} \times \frac{1.6 \times 10^{-3}}{0.8\pi}$$

$$= \frac{4\pi \times 1.6}{0.8 \times 0.8\pi} = \frac{6.4}{0.64}$$

$$= 10 \text{ tesla}$$

Q.23 Answer:

—

Solution:

As the charge remain constant, hence

$Q = CV = C_1 V_1$

$Q = \frac{\varepsilon_0 A}{d} V = \frac{\varepsilon_0 A}{d - t + \frac{t}{K}} V_1$

$\frac{\varepsilon_0 A}{0.005}(25) = \frac{\varepsilon_0 A}{0.002 + \frac{0.003}{10}}(V)$

$V = 11.5 \text{ V}$

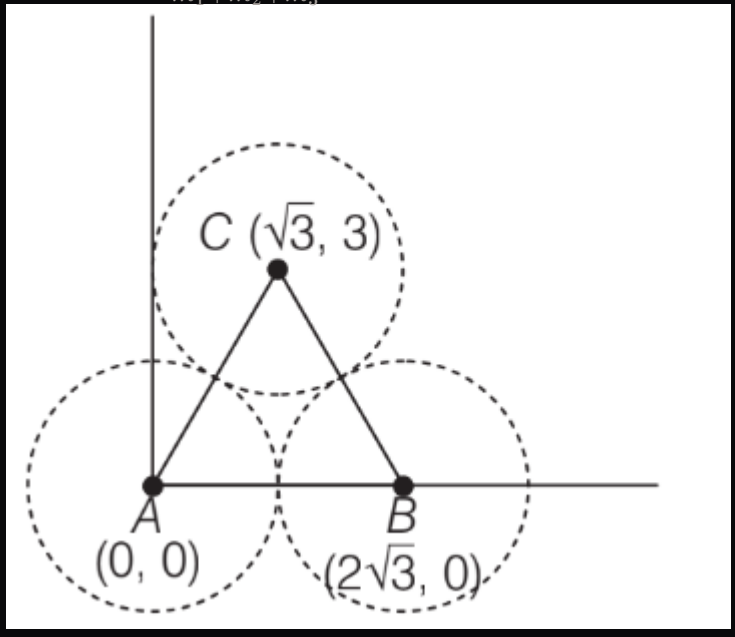
Q.24 Answer:

—

Solution:

The centre of mass of the spheres is given by

$x_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$



As the spheres are identical,

$$\Rightarrow \quad x_{\mathrm{CM}} = \frac{m}{3m}(0 + 2\sqrt{3} + \sqrt{3})$$

$$= \frac{3\sqrt{3}}{3} = \sqrt{3}$$

Similarly,

$$y_{\mathrm{CM}} = \frac{m}{3m}(y_1 + y_2 + y_3)$$

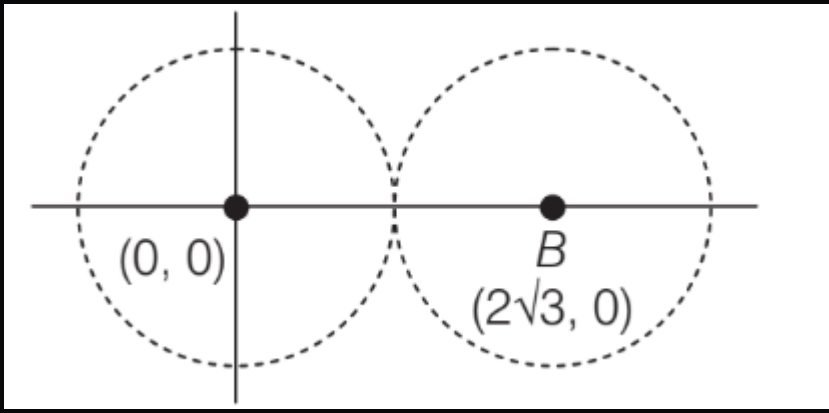
$$= \frac{(0 + 0 + 3)}{3} = 1$$

So, the centre of mass, $C_{\mathrm{CM}} = (\sqrt{3}, 1)$

If one sphere is removed (say C), then

$$x'_{\mathrm{CM}} = \frac{0 + 2\sqrt{3}}{2} = \sqrt{3}$$

$$y'_{\mathrm{CM}} = 0$$



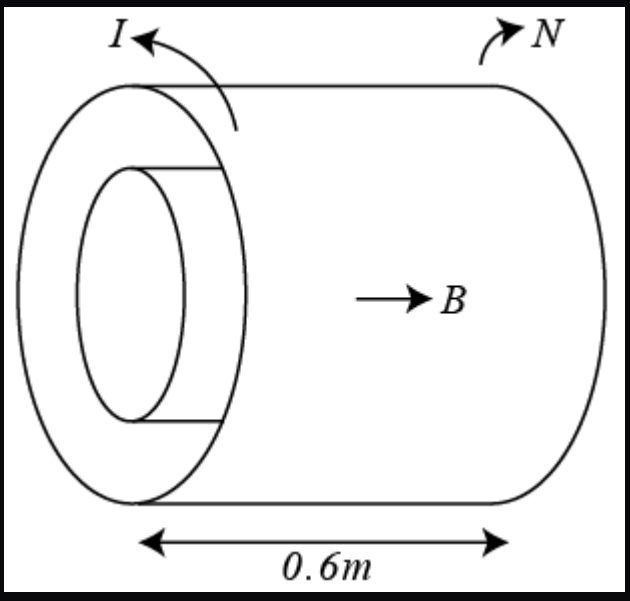
So, $\quad C'_{\mathrm{CM}} = (\sqrt{3}, 0)$

Hence, the centre of mass shifted by $1\ \mathrm{m}$ in $-y$ direction. The correct option is (b).

Q.25 Answer:

—

Solution:



$$
\begin{aligned}
N &= 15/\text{cm} = 1500 \\
B &= \mu_0 N I \\
&= \mu_0 (1500) I \\
\phi &= (B)\left(2 \times 10^{-3}\right)(40 \times 60) \\
&= \underbrace{\mu_0 (1500)\left(2 \times 10^{-3} \times 2.400\right)}_{M} I \propto I \\
M &= 4\pi \times 10^{-7} \times 1.5 \times 10^{3} \times 2 \times 2.4 \\
&= 9.04 \times 10^{-3}\text{H}
\end{aligned}
$$

Q.26 Answer:

$\mathrm{I_3^-}$, $\mathrm{BeCl_2}$

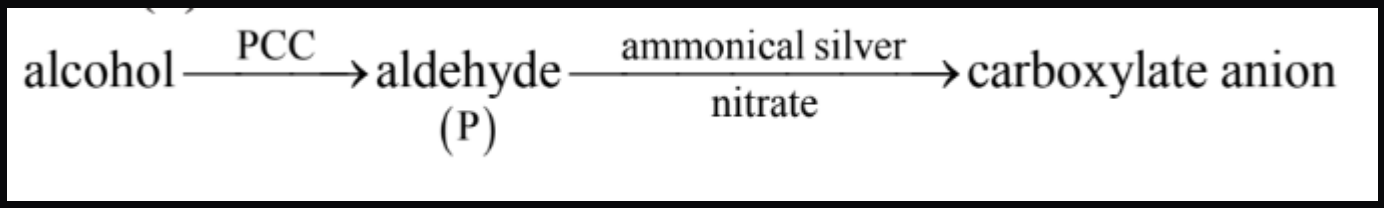
Solution:

Central I in $\mathrm{I_3^-}$ undergoes sp^3d hybridisation giving a linear shape with three lone pairs at equatorial positions. Beryllium chloride has linear structure with sp hybridisation of Be atom.

Q.27 Answer:

Carboxylate anion

Solution:



Q.28 Answer:

C, D

Solution:

Mixing of ideal materials is regarded as an increase in randomness at a molecular level. Therefore, $\Delta S = +$ ve system required for ideal mixing of solution but no enthalpy change present during the formation of ideal solution.

Gibbs free energy of mixing is always negative, thus, ideal solution becomes spontaneous.

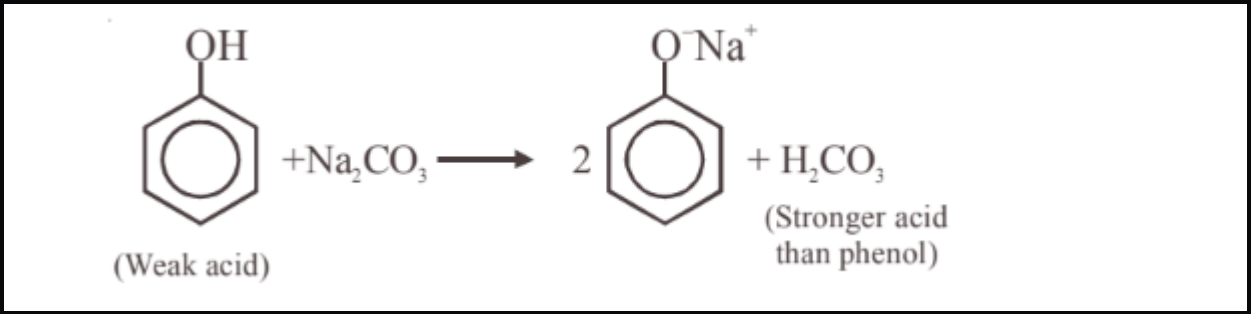
Hence, option (1) is correct.

Q.29 Answer:

Phenol liberates $\mathrm{CO_2}$ from $\mathrm{Na_2CO_3}$ soln

Solution:

Phenol does not liberate CO_2 from Na_2CO_3 solution



Q.30 Answer:

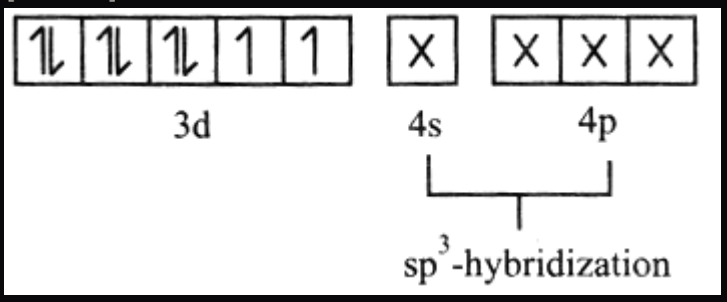
i, iii only

Solution:

$[NiCl_4]^{2-}$ has Ni^{2+} that has the configuration $[Ar]3\,d^8$.

Since Cl^- is a weak-field ligand, the configuration of the complex will be:-

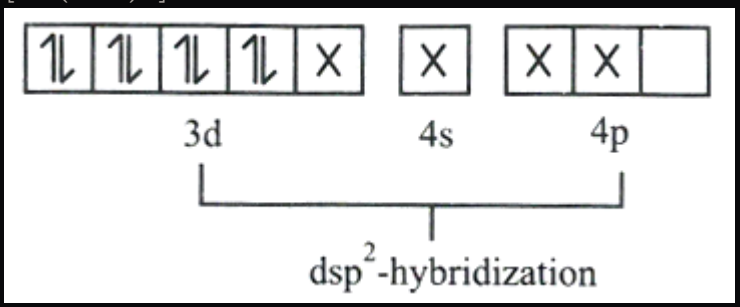
$[NiCl_4]^{2-} =$



Thus, it is a tetrahedral complex.

It is paramagnetic due to unpaired electrons. $[Ni(CO)_4]$ has neutral and strong-field ligand CO. Thus,

$[Ni(CO)_4] =$



Thus, $[Ni(CO)_4]$ is a square-planar and diamagnetic complex.

Therefore, (ii) is incorrect and only (i) and (iii) are correct.

Q.31 Answer:

$Fe_4\,[Fe(CN)_6]_3$

Solution:

Ferric salts (such as $FeCl_3$) form Prussian blue (blue ppt or colouration) with potassium ferrocyanide.

$$4FeCl_3 + 3\,K_4\,[Fe(CN)_6] \longrightarrow \underset{\text{Prussian blue (ferric ferrocyanide)}}{Fe_4\,[Fe(CN)_6]_3} + 12KCl$$

Q.32 Answer:

0.8

Solution:

$$R_f = \frac{\text{distance travelled by solute}}{\text{distance travelled by solvent}}$$
$$R_f = \frac{4}{5} = 0.8$$

Q.33 Answer:

A is incorrect but R is correct

Solution:

The pairing of the two helices of DNA is such that A pairs with T and G pairs with C, through hydrogen bonding.

Thus, number of A = number of T

number of G = number of C

Therefore, statement in Assertion (A) is wrong and Reason (R) is correct.

Q.34 Answer:

$Ca^{2+} < K^+ < Ar$

Solution:

In isoelectronic species atomic radius $\propto \frac{1}{Z_{eff}} \propto \frac{1}{\text{charge on cation}}$

$M > M^+ > M^{2+}$

Hence, increasing order of radius is $Ca^{2+} < K^+ < Ar$ **.**

Q.35 Answer:

Bi_2O_3

Solution:

$SeO_2 \rightarrow$ acidic oxide

$Al_2O_3 \rightarrow$ amphoteric

$Sb_2O_3 \rightarrow$ amphoteric

$Bi_2O_3 \rightarrow$ basic oxide

Q.36 Answer:

p-nitrophenol

Solution:

Steam distillation is the simplest method, which is used to isolate materials that are temperature sensitive like organic compounds. It is applicable at a low boiling point in order to avoid decomposition.

It cannot be applicable in the purification of those organic compounds that are water soluble, non-volatile in steam, vapour pressure low and contains volatile impurities. p-nitrophenol is steam volatile, thus it cannot be used for purifying by steam distillation.

Q.37 Answer:

(1) and (3)

Solution:

Given,

Atomic number of element $B = Z$

($\because$ **noble gas** $\therefore$ **belong to zero group)**

Atomic number of element $A = Z - 1$

(ie, halogens)

Atomic number of element $C = Z + 1$

(ie, group IA)

Atomic number of element $D = Z + 2$

(ie, group IIA)

$\because$ **Element** B **is a noble gas**

$\therefore$ **Element** A **must be a halogen ie, have highest electron affinity.**

and element C **must be an alkali metal and exist in +1 oxidation state.**

and element D **must be an alkaline earth metal with +2 oxidation state.**

Q.38 Answer:

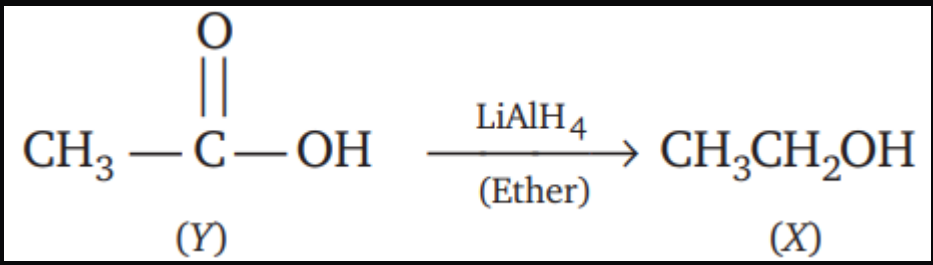
C_2H_5OH, CH_3COOH

Solution:

When ethyl alcohol is oxidised by acidified potassium dichromate, $CH_3COOH(Y)$ **is obtained as :**

$$3C_2H_5OH(X) + 2\ K_2Cr_2O_7 + 8H_2SO_4 \longrightarrow 3CH_3COOH(Y) + 2Cr_2\ (SO_4)_3 + 2\ K_2SO_4 + 11H_2O$$

Carboxylic acid undergo reduction with $LiAlH_4$ **to give primary alcohol as**

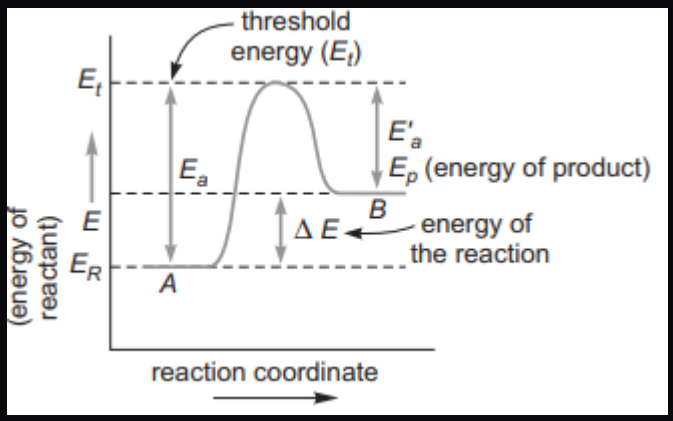


So, X **is** CH_3CH_2OH **and** Y **is** CH_3COOH

Q.39 Answer:

The threshold energy is less than that of activation energy

Solution:



where,

E_a = **activation energy of forward reaction**

E'_a = **activation energy of backward reaction**

The above energy profile diagram shows that

$E_a > E'_a$

The potential energy of the product is greater than that of the reactant, so the reaction is endothermic.

$E_a = E'_a + \Delta E$

$E_t = E_a$ or $E_t > E'_a$

Q.40 Answer:

$3.9\ \text{cm}^{-1}$

Solution:

Conductivity $(\kappa) = \frac{1}{\text{Resistance } (R)} \times$ **cell constant Cell constant** $= \kappa \times R$

Given, $\kappa = 0.0135\ \text{cm}^{-1} \Rightarrow R = 300\Omega$

Cell constant $= 0.0135 \times 300 = 3.9\ \text{cm}^{-1}$.

Q.41 Answer:

$4 \times 10^2\ \text{atm}$

Solution:

$$\text{SO}_2 + \frac{1}{2}\text{O}_2 \rightleftharpoons \text{SO}_3$$

$$K_1 = \frac{[\text{SO}_3]}{[\text{SO}_2]\,[\text{O}_2]^{1/2}}$$

$$2\text{SO}_3 \rightleftharpoons 2\text{SO}_2 + \text{O}_2$$

$$\text{K}_2 = \frac{[\text{SO}_2]^2\,[\text{O}_2]}{[\text{SO}_3]^2}$$

From Eqs. (i) and (ii)

$$K_2 = \frac{1}{K_1^2}$$

$$K_2 = \frac{1}{(5 \times 10^{-2})^2} = \frac{1}{25 \times 10^{-4}}$$

$$= \frac{100 \times 10^2}{25}$$

$$= 4 \times 10^2 \text{ atm}$$

Q.42 Answer:

III V I II

Solution:

(A) Resonance-Ben zene shows the resonance.

(B) Inductive effect 1-chloropropane shows the highest inductive effect.

(C) Electromeric effect Ethene shows the electromeric effect.

(D) Hyper conjugation Eth ane shows the hyper conjugation in these species.

Q.43 Answer:

$$C_2H_2 + H_2O \xrightarrow[1\%HgSO_4 60^\circ C]{40\%H_2SO_4}$$

Solution:

$$C_2H_2 + H_2O \xrightarrow[1\%\ HgSO_4]{40\%\ H_2SO_2} \underset{\text{Acetaldehyde}}{CH_3CHO}$$

Acetaldehyde + Fehling's solution $\xrightarrow{\Delta}$ Cuprous oxide (Red ppt.)

Q.44 Answer:

(ii), (iv)

Solution:

$$Fe^{2+} + 2e^- \longrightarrow Fe; E^\circ = -0.44$$

$$Cu^{2+} + 2e^- \longrightarrow Cu; E^\circ = +0.34$$

$$Ag + 1e^- \longrightarrow Ag; E^\circ = +0.80$$

If a metal have negative electrode potential than metal easily displace, other metal, i.e. Fe can displace Cu and Ag.

Q.45 Answer:

2-bromo-3-hydroxy benzonitrile

Solution:



The priority order for the given functionalities in

$-\mathrm{CN} > -\mathrm{OH} > -\mathrm{Br}.$

$\therefore$ The numbering will be done from $-\mathrm{CN}$ and in such a way that the groups get lowest number possible.

Hence, the IUPAC name will be

2-bromo-3-hydroxy benzonitrile.

Q.46 Answer:

—

Solution:

$\mathrm{H}_2^+ = 1e^-$	ie. σls^1	$BO = \frac{1-0}{2} = 0.5$
$\mathrm{He}_2^+ = 3e^-$	ie. $\sigma/s^2\sigma^* ls'$	$BO = \frac{2-1}{2} = 0.5$
$\mathrm{He}_2^- = 5e^-$	ie. $\sigma/s^2\sigma^*/s^2\sigma 2s'$	$BO = \frac{3-2}{2} = 0.5$
$B_2^+ = 9e^-$		$BO = \frac{N-8}{2} = \frac{9-8}{2} = 0.5$
$\mathrm{F}_2^- = 19e^-$		$BO = \frac{20-\mathrm{N}}{2} = \frac{20-19}{2} = 0.5$
$Be_2^{2-} = 10e^-$		$BO = \frac{10-8}{2} = 1$

Q.47 Answer:

—

Solution:

$\mathrm{Pr}, \mathrm{Nd}, \mathrm{Tb}$ **and Dy exhibit +4 state in their oxides as they acquire configurations closer to stable** $4\mathrm{f}^\circ$ **and** $4\mathrm{f}^7$ **configurations.**

$\mathrm{Pr}^{4+} = [\mathrm{Xe}]4\mathrm{f}^1, \mathrm{Nd}^{4+} = [\mathrm{Xe}]4\mathrm{f}^2, \mathrm{Pm}^{4+} = [\mathrm{Xe}]4\mathrm{f}^3$

$\mathrm{Sm}^{4+} = [\mathrm{Xe}]4\mathrm{f}^4, \mathrm{Eu}^{4+} = [\mathrm{Xe}]4\mathrm{f}^5, \mathrm{Gd}^{4+} = [\mathrm{Xe}]4\mathrm{f}^6,$

$\mathrm{Tb}^{4+} = [\mathrm{Xe}]4\mathrm{f}^7, \mathrm{Dy}^{4+} = [\mathrm{Xe}]4\mathrm{f}^6.$

Q.48 Answer:

—

Solution:

Step I : Calculate volume

$= \text{ area } \times \text{ thickness (two times)}$

$= 1 \times 2\ \mathrm{cm}^2 \times 2 \times 0.01 = 0.04$

Step II : Mass of material deposited

$= 0.04 \times 9 = 0.36\ \mathrm{g} = \text{ volume } \times \text{ density}$

Step III : According to Faraday law,

$$m = Zit$$

$$i = 1.5 \text{ A}$$

$$Z = 0.0003 \text{ g/C}$$

$$t = \frac{m}{Zi} = \frac{0.36}{0.0003 \times 1.5}$$

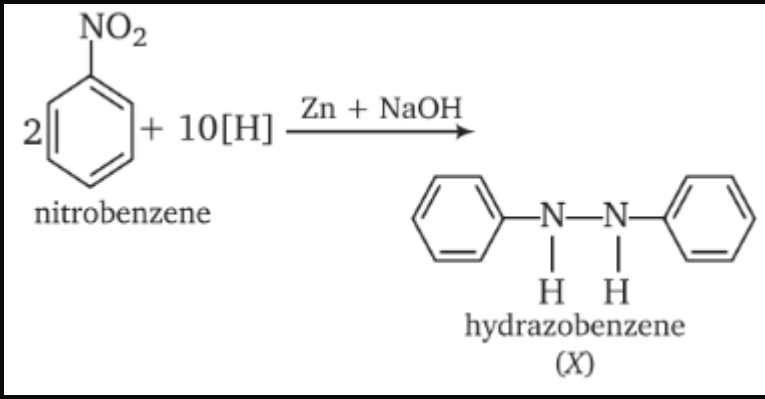
$$t = 800 \text{ s}$$

Q.49 Answer:

—

Solution:

Nitrobenzene on reduction using zinc in alkaline medium gives hydrazobenzene.



Hydrazobenzene (X) contains 27σ and 6π bonds.

Q.50 Answer:

—

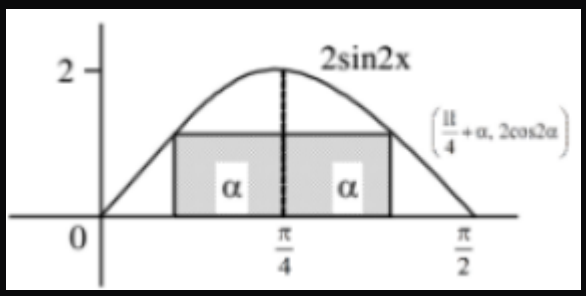
Solution:

As there are 3 subshells, those will be s, p and d giving a total of $= 1s^2, 2s^2, 2p^2, 3s^2, 3p^6, 3d^{10} = 28$ electrons and 1 electrons for each orbital with $m_s = -1/2$ and $+1/2$ each. Thus, there will be 14 such electrons.

Q.51 Answer:

$\frac{\pi}{2\sqrt{3}}$

Solution:



$$\text{Perimeter } = 2(2\alpha + 2\cos 2\alpha)$$
$$\mathrm{P} = 4(\alpha + \cos 2\alpha)$$
$$\frac{\mathrm{dP}}{\mathrm{d}\alpha} = 4(1 - 2\sin 2\alpha) = 0$$
$$\sin 2\alpha = \frac{1}{2}$$
$$2\alpha = \frac{\pi}{6}, \frac{5\pi}{6}$$
$$\frac{\mathrm{d}^2\mathrm{P}}{\mathrm{d}\alpha^2} = -4\cos 2\alpha$$
$$\text{for maximum } \alpha = \frac{\pi}{12}$$
$$\text{Area } = (2\alpha)(2\cos 2\alpha)$$
$$= \frac{\pi}{6} \times 2 \times \frac{\sqrt{3}}{2} = \frac{\pi}{2\sqrt{3}}$$

Q.52 Answer:

0.25

Solution:

$$\mathrm{g}(f(\mathrm{x})) = \mathrm{x} \Rightarrow \mathrm{g}'(f(\mathrm{x})) \cdot f'(\mathrm{x}) = 1$$
$$f(\mathrm{x}) = 1 \Rightarrow \mathrm{x} = 0$$
$$\mathrm{g}'(1).f'(0) = 1$$
$$f'(\mathrm{x}) = 2\mathrm{x} + \mathrm{e}^{-\mathrm{x}}$$
$$f'(0) = 1$$
$$\mathrm{g}'(1) = 1$$

Q.53 Answer:

$\frac{59}{99}$

Solution:

Total number of ways in which $n_1 + n_2 = 100$ is equal to 99.

Now, $\mathrm{n}_1 \cdot \mathrm{n}_2 > 1600$

$$\Rightarrow \mathrm{n}_1 (100 - \mathrm{n}_1) > 1600$$
$$\Rightarrow \mathrm{n}_1^2 - 100\mathrm{n} + 1600 < 0$$
$$\Rightarrow (\mathrm{n}_1 - 80)(\mathrm{n}_1 - 20) < 0$$
$$\Rightarrow 20 < \mathrm{n}_1 < 80$$
$$\Rightarrow 21 \leq \mathrm{n}_1 \leq 79$$

Thus, number of favorable ways $= 79 - 21 + 1 = 59$

Hence, required probability $= \frac{59}{99}$.

Q.54 Answer:

286

Solution:

$$3n-2=7m+2$$
$$n=\frac{7m+4}{3}$$
$$n=\frac{7(3\lambda+2)+4}{3}$$
$$n=7\lambda+6 \text{ and } m=3\lambda+2$$
$$3\lambda+2\le 2004 \text{ and } 7\lambda+6\le 2004$$
$$\lambda\le 667 \text{ and } \lambda\le 285$$
$$\Rightarrow 286 \text{ common terms}$$

Q.55 Answer:

3

Solution:

Let $x^2-5x+4=a \quad \ldots(1)$

$\therefore \quad a\left(y^2+y+1\right)<2y$

$\Rightarrow \quad ay^2+y(a-2)+a<0$ for all $y\in R$

$\therefore \quad a<0$ and $D<0$

$\Rightarrow \quad a<0$ and $(a-2)^2-4a^2<0$

$\Rightarrow \quad 3a^2+4a-4>0$

$\therefore \quad a<0$ and $a<-2$

$\Rightarrow \quad a<-2 \quad \ldots$ (ii)

From Eqs. (i) and (ii), $x^2-5x+4<-2$

$\Rightarrow x^2-5x+6<0 \Rightarrow x\in(2,3)$

Q.56 Answer:

8

Solution:

$$\lim_{x\to\frac{\pi}{2}}\frac{4\sqrt{2}\cdot 2\sin 2x\cos x}{2\sin 2x\sin\frac{3x}{2}+\left(\cos\frac{5x}{2}-\cos\frac{3x}{2}\right)-\sqrt{2}(1+\cos 2x)}$$
$$\lim_{x\to\frac{\pi}{2}}\frac{16\sqrt{2}\sin x\cos^2 x}{2\sin 2x\left(\sin\frac{3x}{2}-\sin\frac{x}{2}\right)-2\sqrt{2}\cos^2 x}$$
$$\lim_{x\to\frac{\pi}{2}}\frac{16\sqrt{2}\sin x\cos^2 x}{4\sin x\cos x\left(2\cos x\cdot\sin\frac{x}{2}\right)-2\sqrt{2}\cos^2 x}$$
$$\frac{16\sqrt{2}\sin x}{8\sin x\cdot\sin\frac{x}{2}-2\sqrt{2}}=8$$

Q.57 Answer:

atleast 10

Solution:

Given, $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

For unique solution, $\det(A) \neq 0$

Case I

$$\det(A) = \begin{vmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{vmatrix} = 2abc \neq 0$$

Here, a, b, c **is selected from** $1, 0, 0$. **[no case is possible]**

Case II

(i) $$\det(A) = \begin{vmatrix} 1 & a & b \\ a & 0 & c \\ b & c & 0 \end{vmatrix} = 2abc - c^2 \neq 0$$

Here, a, b, c **are selected from** $1, 1, 0$. **[2 cases are possible]**

(ii) $$\det(A) = \begin{vmatrix} 0 & a & b \\ a & 1 & c \\ b & c & 0 \end{vmatrix} = 2abc - b^2 \neq 0$$

Here, a, b, c **are selected from** $1, 1, 0$. **[2 cases are possible]**

(iii) $$\det(A) = \begin{vmatrix} 0 & a & b \\ a & 0 & c \\ b & c & 1 \end{vmatrix}$$

$2abc - a^2 \neq 0$

There is no constraint on a,b,c

infinite matrices can be there.

Q.58 Answer:

$\frac{4\sqrt{13}}{5}$ units

Solution:

$P = (-4, 0) \quad Q = (0, 6)$

Let $\mathrm{A} = (\mathrm{x}, \mathrm{y})$

$$\frac{\mathrm{PA}}{\mathrm{QA}} = \frac{2}{3} \Rightarrow 9\mathrm{PA}^2 = 4\mathrm{QA}^2$$

$$9\left[(\mathrm{x}+4)^2 + \mathrm{y}^2\right] = 4\left[\mathrm{x}^2 + (\mathrm{y}-6)^2\right]$$

$$5\mathrm{x}^2 + 5\mathrm{y}^2 + 72\mathrm{x} + 48\mathrm{y} = 0$$

is equation of circum circle of $\triangle \mathrm{ABC}$

Circumcentre of $\triangle \mathrm{ABC} = \mathrm{S} = \left(-\frac{36}{5}, -\frac{24}{5}\right)$

and A is orthocenter

$\mathrm{SG}:\mathrm{GA}=1:2$

$\Rightarrow \mathrm{SG}=\frac{1}{3}\mathrm{SA}=\frac{1}{3}\sqrt{\left(\frac{36}{5}\right)^2+\left(\frac{24}{5}\right)^2}=\frac{4\sqrt{13}}{5}$ units

Q.59 Answer:

8

Solution:

$\Delta=0 \& a_0+a_1+a_2\neq 0$

then $a_0=a_1=a_2$

$\Delta=-\left[\left(a_0^3+a_1^3+a_2^3\right)-3a_0a_1a_2\right]$

putting $x=0$

$a_0=1$

coefficient of $x=4a$

coefficient of $x^2=4b+6a^2$

$1=4a=6a^2+4b$

$\Rightarrow a=\frac{1}{4}; b=\frac{5}{32} 5\times\left(\frac{1/4}{5/32}\right)=8$

Q.60 Answer:

2

Solution:

We have, $xdx=\frac{dy}{y}-\frac{dy}{1+y}$

∴ On integration, we get

$\mathrm{C}+\frac{\mathrm{x}^2}{2}=\ln \mathrm{y}-\ln(1+\mathrm{y})$

As $f(0)=1\Rightarrow \mathrm{C}=-$ **åŠ¨ 2**

$\Rightarrow \frac{2y}{1+y}=e^{\frac{x^2}{2}}$

$\therefore 2f(\mathrm{x})=\left(1+(f(\mathrm{x}))\mathrm{e}^{\frac{x^2}{2}}\right.$

Put $\mathrm{x}=2$**, we get** $2f(2)=(1+f(2))\mathrm{e}^2$

$\therefore \quad k=2$

Q.61 Answer:

17.1

Solution:

Let $g'(0)=a, g''(1)=b$

$\therefore \quad f(x)=x^2+ax+b$

$f'(x)=2x+a$

$f'(x)=2$

$\therefore \quad g(y)=y^2+2+(2y+a)-2$

$g(y)=y^2+2y+a$

$g'(y)=2y+2 \Rightarrow g'(0)=2=a$

$g''(y)=2 \Rightarrow g''(1)=2=b$

$\therefore \quad f(x)=x^2+2x+2 \quad a=2,\ b=2$

$g(y)=y^2+2y+2$

$\therefore \quad f(3)+\frac{1}{g(2)}=17+\frac{1}{10}=17.1$

Q.62 Answer:

4

Solution:

$$AS_1+AS_2 \geq 2\sqrt{AS_1 \cdot AS_2}$$

$$4 \geq 2\sqrt{AS_1 \cdot AS_2}$$

$$AS_1 \cdot AS_2 \leq 4$$

Q.63 Answer:

-1

Solution:

$\left(x^2+2\right)^2-(a+4)\left(x^2+2\right)\left(x^2+1\right)+(a+3)\left(x^2+1\right)^2=0$

Divide by $\left(x^2+1\right)^2$ **:**

$\left(\frac{x^2+2}{x^2+1}\right)^2-(a+4)\left(\frac{x^2+2}{x^2+1}\right)+a+3=0$

let $\frac{x^2+2}{x^2+1}=t$

$\therefore t^2-(a+4)t+(a+3)=0$

$\Rightarrow t=1; t=a+3$

for $t=1:\ \frac{x^2+2}{x^2+1}=1 \Rightarrow x^2+2=x^2+1$

Not possible

for $t=a+3:\ \frac{x^2+2}{x^2+1}=a+3$

$\Rightarrow a+3=\underbrace{1+\frac{1}{x^2+1}}_{\text{Range: (1,2]}}$

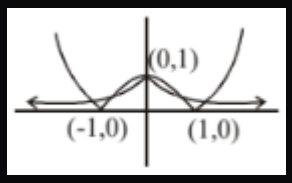
$\therefore$ **for atleast one solution :** $1<a+3\leq 2$

$\Rightarrow -2<a\leq -1$

Q.64 Answer:

5

Solution:



$\lim_{a \to x} \frac{e^{|x|}\left(e^{a^2-x^2}-1\right)}{\frac{(a+x)\sin(a-x)\cdot(a-x)}{(a-x)}} = \mathrm{e}^{|x|}$

Now $f(x)\left(\left|x^2-1\right|\right)=1 \Rightarrow \left|x^2-1\right|=e^{-|x|}$

Q.65 Answer:

can be $\sqrt{80}$

Solution:

$$\text{The shortest distance (SD)} = \frac{\begin{vmatrix} 2-1 & 4-2 & 5-3 \\ 2 & 3 & 4 \\ 3 & 2 & 5 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 2 & 5 \end{vmatrix}} = \frac{1}{\sqrt{78}},$$

$$\text{So, } \frac{1}{\text{SD}} = \sqrt{78}$$

Q.66 Answer:

1

Solution:

$$\text{Given, } \frac{\sin^{-1} x^2 + \cos^{-1} x}{\cos^{-1} x^2 + \sin^{-1} x} = -3$$

$$\Rightarrow \sin^{-1} x^2 + \cos^{-1} x = -3\cos^{-1} x^2 - 3\sin^{-1} x$$

$$\Rightarrow \quad \sin^{-1} x^2 + 3\cos^{-1} x^2 + \cos^{-1} x + 3\sin^{-1}$$

$$x = 0 \Rightarrow \pi + 2\cos^{-1} x^2 + 2\sin^{-1} x = 0$$

$$\Rightarrow \quad \cos^{-1} x^2 + \sin^{-1} x = \frac{-\pi}{2}$$

Since, $0 \leq \cos^{-1} x^2 \leq \frac{\pi}{2}$ **and**

$\frac{-\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$

So, $\cos^{-1} x^2 = 0 \Rightarrow x = \pm 1$ **and**

$\sin^{-1} x = \frac{-\pi}{2} \Rightarrow x = -1.$

∴ Only solution is $\mathrm{x} = \alpha = -1.$

Hence, $\frac{\sec^{-1}\alpha - \tan^{-1}\alpha}{\cot^{-1}\alpha - \operatorname{cosec}^{-1}\alpha} =$

$\frac{\sec^{-1}(-1) - \tan^{-1}(-1)}{\cot^{-1}(-1) - \operatorname{cosec}^{-1}(-1)} = \frac{\pi + \frac{\pi}{4}}{\frac{3\pi}{4} + \frac{\pi}{2}} = \frac{\frac{5\pi}{4}}{\frac{5\pi}{4}}$

Q.67 Answer:

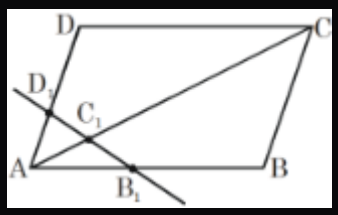
$\frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

Solution:

$\overrightarrow{AB} = \lambda_1 \overrightarrow{AB}$

$\overrightarrow{AD_1} = \lambda_2 \overrightarrow{AD}$

$\overrightarrow{AC_1} = \lambda_3 \overrightarrow{AC}$



As, $ABCD$ is a parallelogram

$\Rightarrow \overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$

$\Rightarrow \overrightarrow{AB} + \overrightarrow{AD} = \overrightarrow{AC}$

$\Rightarrow \dfrac{\overrightarrow{AB_1}}{\lambda_1} + \dfrac{\overrightarrow{AD_1}}{\lambda_2} = \dfrac{\overrightarrow{AC_1}}{\lambda_3}$

$\Rightarrow \dfrac{\overrightarrow{AB_1}}{\lambda_1} + \dfrac{1}{\lambda_2}\overrightarrow{AD_1} + \left(\dfrac{-1}{\lambda_3}\right)\overrightarrow{AC_1} = \overrightarrow{0}$

Here points B, C and D are collinear

$\Rightarrow \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{-1}{\lambda_3} = 0 \Rightarrow \frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

Q.68 Answer:

$c = 20$

Solution:

Length of the chord $= 3\sqrt{\frac{49}{9} + \frac{49}{9}} = 7\sqrt{2}$

$\text{Radius } = \sqrt{\frac{49}{2} + \frac{1}{2}} = 5$

$(5, 5)$ **point lies on a circle**

$25 + 25 - 4 \times 5 - 2 \times 5 = C = 20$

Q.69 Answer:

$1/7$

Solution:

H: tossing a Head, $P(H) = \frac{1}{2}$; **A : event of tossing a 2 with die,** $P(A) = \frac{1}{6}$ **E: tossing a 2 before tossing a head**

$P(E) = P(\bar{H} \cap A \text{ or } \{(\bar{H} \cap \bar{A}) \text{ and } \bar{H} \cap A\} \text{ or } \ldots\ldots)$

$= \left(\dfrac{1}{2} \cdot \dfrac{1}{6}\right) + \left(\dfrac{1}{2} \cdot \dfrac{5}{6}\right) \cdot \left(\dfrac{1}{2} \cdot \dfrac{1}{6}\right) + \ldots\ldots = \dfrac{1}{12} + \dfrac{5}{12} \cdot \dfrac{1}{12} + \ldots\ldots\infty$

$P(E) = \dfrac{\frac{1}{12}}{1 - \frac{5}{12}} = \dfrac{1}{7}$

Q.70 Answer:

-2.25

Solution:

$$\frac{dy}{dx} = \frac{y}{x} + x^2$$
$$\frac{dy}{dx} + \left(-\frac{1}{x}\right) y = x^2$$
$$\text{If } = e^{\int -\frac{1}{x} dx} \Rightarrow IF = \frac{1}{x}$$
$$\therefore y\left(\frac{1}{x}\right) = \int x^2 \cdot \frac{1}{x} dx = \frac{x^2}{2} + c$$
$$\Rightarrow y = \frac{x^3}{2} + cx$$
$$\text{pass thru } (1, -2) \Rightarrow c = -\frac{5}{2}$$
$$\therefore y = \frac{x^3}{2} - \frac{5x}{2}$$
$$\therefore 2\int_0^1 y dx = 2 \cdot \int_0^1 \left(\frac{x^3}{2} - \frac{5x}{2}\right) dx = -2.25$$

Q.71 Answer:

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Solution:

$$\lim_{t \to 0} \frac{\int_0^t \left(\frac{1-\cos(3x)}{x^2+3}\right) dx}{t^3} dx \quad \left(\frac{0}{0} \text{ form}\right)$$
$$= \lim_{t \to 0} \left(\frac{1 - \cos(3t)}{t^2 + 3}\right) \times \frac{1}{3t^2} = \frac{1}{2}$$
$$\text{and } \int_{\frac{1}{2}}^{\frac{1}{2}} \sin\left(\tan^{-1} \frac{x^2}{\sqrt{1 - x^4}}\right) dx$$
$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} \sin\left(\sin^{-1} x^2\right) dx$$
$$= \int_{-1/2}^{1/2} \left(x^2\right) dx = 2\int_0^{1/2} x^2 dx = \frac{1}{12}$$

Q.72 Answer:

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Solution:

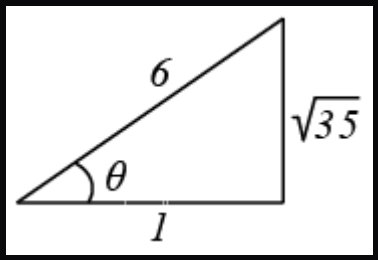
$$|\vec{a}+\vec{b}|=|-\vec{c}-\vec{d}|$$
$$\Rightarrow \frac{1}{4}+\frac{1}{9}+2\cdot\frac{1}{2}\cdot\frac{1}{3}\cos\theta_1=\frac{1}{36}+1+2\cdot\frac{1}{6}\cdot 1\cos\theta_2$$
$$\Rightarrow \cos\theta_1-\cos\theta_2=2$$

Q.73 Answer:

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Solution:

$$m=-(3l+5n)^2$$
$$-6(3l+5n)n-2nl-5l(3l+5n)=0$$
$$-18\ln-30n^2-2nl-15l^2-25ln=0$$
$$15l^2+45nl+30n^2=0$$
$$5l^2+15nl+10n^2=0$$
$$l^2+3nl+2n^2=0$$
$$(l+2n)(l+n)=0$$
$$l=-2n|l=-n$$
$$m=n|m=-2n$$
$$(-2n,n,n)|(-n,-2n,n)$$
$$n_1\to(-2,1,1)|n_2\to(-1,-2,1)$$
$$\cos\theta=\frac{\vec{n}_1\cdot\vec{n}_2}{|n_1|\cdot|n_2|}=\frac{\not{2}-2+1}{\left(\sqrt{2^2+1+1}\right)^2}=\frac{1}{6}$$
$$\left[\frac{6\tan^2\theta-5\sec\theta}{10\pi}\right]=\left[\frac{6\times 35-5\times 6}{10\pi}\right]$$
$$=\left[\frac{18\phi}{10\pi}\right]=5$$



Q.74 Answer:

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Solution:

Iterating F we get:

$$F(z) = \frac{z+i}{z-i}$$

$$F(F(z)) = \frac{\frac{z+i}{z-i}+i}{\frac{z+i}{z-i}-i} = \frac{(z+i)+i(z-i)}{(z+i)-i(z-i)} = \frac{z+i+zi+1}{z+i-zi-1} = \frac{(z+1)(i+1)}{(z-1)(1-i)}$$

$$= \frac{(z+1)(i+1)^2}{(z-1)\left(1^2+1^2\right)} = \frac{(z+1)(2i)}{(z-1)(2)} = \frac{z+1}{z-1}i$$

$$F(F(F(z))) = \frac{\frac{z+1}{z-1}i+i}{\frac{z+1}{z-1}i-i} = \frac{\frac{z+1}{z-1}+1}{\frac{z+1}{z-1}-1} = \frac{(z+1)+(z-1)}{(z+1)-(z-1)} = \frac{2z}{2} = z$$

From this, it follows that $z_{k+3} = z_k$, for all k.

Thus $z_{2002} = z_{3\cdot 667+1} = z_1 = \frac{z_0+i}{z_0-i} = \frac{\frac{1}{137}+2i}{\frac{1}{137}} = 1+274i$

Thus $a+b = 1+274 = 275.$

Q.75 Answer:

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Solution:

Let D denotes dancing. P denotes painting and S denotes singing.

$\therefore \quad n(D \cup P \cup S) = 265$

$n(S) = 200, n(D) = 110, n(P) = 55$

$n(S \cap D) = 60, n(S \cap P) = 30$

and $n(D \cap P \cap S) = 10$

$\because n(D \cup P \cup S) = n(D) + n(P) + n(S) - n(D \cap P)$

$- n(P \cap S) - n(S \cap D) + n(D \cap P \cap S)$

$\therefore \quad 265 = 110 + 55 + 200 - n(D \cap P)$

$\Rightarrow \quad 265 = 285 - n(D \cap P)$

$\Rightarrow \quad n(D \cap P) = 20$

$\therefore$ Only person who like dancing and painting

$= n(D \cap P) - n(D \cap P \cap S)$

$= 20 - 10 = 10$